



1000 AI TERMS

EVERY DATA PROFESSIONAL SHOULD KNOW

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- Algorithm: Step-by-step procedure to solve a problem.
- Artificial Intelligence (AI): Machines mimicking human intelligence.
- Neural Network: Brain-inspired interconnected nodes for pattern recognition.
- Supervised Learning: Training models with labeled data (e.g., spam detection).
- Unsupervised Learning: Finding patterns in unlabeled data (e.g., clustering).
- Reinforcement Learning: Learning via rewards/penalties (e.g., game AI).
- Bias-Variance Tradeoff: Balancing underfitting and overfitting.
- Feature Engineering: Crafting input variables to improve models.
- Hyperparameter: Configurations set before training (e.g., learning rate).
- Loss Function: Measures model prediction error (e.g., MSE).

- Overfitting: Model memorizes training data, fails on new data.
- Generalization: Model performs well on unseen data.
- Epoch: One full pass through the training dataset.
- Activation Function: Adds non-linearity to neural networks (e.g., ReLU).
- Gradient Descent: Optimizes models by minimizing loss.
- Backpropagation: Adjusts neural network weights via error correction.
- Ensemble Learning: Combines models for better accuracy (e.g., Random Forest).
- Transfer Learning: Reusing pre-trained models for new tasks.
- Model Drift: Performance decline due to data changes over time.
- Explainable AI (XAI): Making model decisions interpretable.

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Machine Learning

- Regression: Predicts continuous values (e.g., house prices).
- Classification: Assigns labels (e.g., fraud/not fraud).
- Clustering: Groups similar data points (e.g., customer segments).
- Decision Tree: Flowchart-like model for decisions.
- Random Forest: Ensemble of decision trees for robustness.
- Gradient Boosting: Sequentially corrects errors (e.g., XGBoost).
- Cross-Validation: Tests model generalizability (e.g., k-fold).
- Precision/Recall: Metrics for classification performance.
- ROC-AUC: Evaluates classifier tradeoffs (TPR vs. FPR).
- Confusion Matrix: Tracks TP, TN, FP, FN.

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Machine Learning

- SVM (Support Vector Machine): Finds optimal hyperplanes for classification.
- K-Means: Groups data into k clusters.
- PCA (Principal Component Analysis): Reduces dimensionality while preserving variance.
- Regularization: Prevents overfitting (e.g., L1/L2).
- Outlier Detection: Identifies anomalies in data.
- Feature Selection: Picks most relevant input variables.
- Dimensionality Reduction: Simplifies data without losing critical info.
- Bagging: Trains models in parallel (e.g., Random Forest).
- Boosting: Trains models sequentially to fix errors (e.g., AdaBoost).
- MLOps: Automates ML lifecycle (deployment, monitoring).

- CNN (Convolutional Neural Network): Processes grid-like data (e.g., images).
- RNN (Recurrent Neural Network): Handles sequences (e.g., time-series).
- Transformer: Architecture for parallel processing (e.g., BERT, GPT).
- Autoencoder: Compresses data, detects anomalies.
- GAN (Generative Adversarial Network): Generates synthetic data (e.g., deepfakes).
- Attention Mechanism: Focuses on relevant input parts (e.g., translation).
- Dropout: Reduces overfitting by deactivating neurons randomly.
- Batch Normalization: Stabilizes training by normalizing layer inputs.
- Learning Rate: Step size in gradient descent.
- LSTM (Long Short-Term Memory): RNN variant for long-term dependencies.

- GRU (Gated Recurrent Unit): Simplified LSTM for efficiency.
- Embedding Layer: Converts categorical data to vectors.
- Vanishing Gradient: Issue where gradients become too small during training.
- YOLO (You Only Look Once): Real-time object detection model.
- Fine-Tuning: Adapts pre-trained models to new tasks.
- Tokenization: Splits text into words/subwords.
- Beam Search: Heuristic for sequence prediction (e.g., text generation).
- Transformer Architecture: Uses self-attention for NLP tasks.
- Self-Attention: Weights input relevance dynamically.
- Weight Initialization: Method to set initial neural network weights.

- Tokenization: Splitting text into words/subwords.
- BERT: Bidirectional language model for context understanding.
- TF-IDF: Scores word importance in documents.
- Word2Vec: Generates word embeddings for semantic analysis.
- Sentiment Analysis: Detects emotion in text (e.g., reviews).
- NER (Named Entity Recognition): Identifies entities (e.g., people, places).
- Stemming/Lemmatization: Reduces words to root forms.
- Seq2Seq: Converts input sequences to output sequences (e.g., translation).
- Masked Language Modeling: Predicts masked words in sentences.
- GPT: Generative language model for text creation.

- Zero-Shot Learning: Predicts unseen classes without training.
- Text Summarization: Condenses long text into key points.
- Topic Modeling: Discovers themes in documents (e.g., LDA).
- POS Tagging: Labels words with grammatical roles (e.g., noun, verb).
- Dependency Parsing: Analyzes grammatical structure.
- Corpus: Large text dataset for training/models.
- Stop Words: Common words excluded from analysis (e.g., "the").
- N-gram: Contiguous sequence of n words.
- Token Embeddings: Numerical representations of tokens.
- BPE (Byte-Pair Encoding): Subword tokenization method.

- Bias: Skewed model outputs due to unfair data.
- Fairness: Ensuring equitable outcomes across groups.
- Explainability: Making AI decisions interpretable.
- GDPR: EU regulation protecting user data privacy.
- Adversarial Attack: Manipulates inputs to fool models.
- Data Anonymization: Protects user identities.
- Model Audit: Evaluates models for compliance/ethics.
- AI Governance: Policies for responsible AI use.
- Transparency: Clear disclosure of AI decision-making.
- Responsible AI: Ethical design/deployment of AI systems.

- Privacy-Preserving AI: Techniques to protect data (e.g., federated learning).
- Federated Learning: Trains models on decentralized data.
- Model Cards: Documents model performance/limitations.
- Ethical AI: Aligns AI with moral principles.
- Data Sovereignty: Laws governing data storage by geography.
- Algorithmic Accountability: Responsibility for AI outcomes.
- AI Safety: Ensures AI systems behave as intended.
- Synthetic Data: Artificially generated data for training.
- Differential Privacy: Protects individual data in datasets.
- Compliance: Adherence to legal/regulatory standards.